

Installation & User Guide: Hosi Noise Suppression

Welcome to **Hosi Noise Suppression** - a professional tool out of the box for REAPER that utilizes the DeepFilterNet AI engine to clean up background noise, remove interference, and achieve studio-quality audio.

This document will guide you from the initial installation steps to advanced usage configurations.

1. System Requirements & Installation

Prerequisites:

1. **REAPER**: The latest version is recommended.
2. **RealmGui**: The graphical user interface library for REAPER scripts. (Can be installed directly via ReaPack in REAPER).
3. **DeepFilterNet (AI Engine)**: The AI processing core written in Python.
 - Install DeepFilterNet via your Python environment running:
 - `pip install deepfilternet`

Script Installation in REAPER:

1. Step 1: Open REAPER.
2. Step 2: Press `?` or go to the menu **Actions > Show action list...**
3. Step 3: In the bottom right corner, select **New action > Load ReaScript...**
4. Step 4: Point to the `Hosi_DeepFilterNet.lua` file on your computer.
5. Step 5: (Optional) Drag the script from the Action List and drop it onto your REAPER Toolbar to create a handy shortcut button.

2. Interface & Core Features

To launch the tool, select the Audio Items on the Timeline that you want to denoise, then run the `Hosi_DeepFilterNet.lua` action.

An elegant Dark Mode control panel named **Hosi Noise Suppression** will appear.

Control Parameters:

- **Post-Filter (Deep Cleaning)**:
 - *Default*: Unchecked.
 - *Effect*: When checked, the AI thoroughly squeezes and removes persistent background noise. Suitable for voice recording signals with too much environmental garbage. However, overusing this may cause vocal artifacts.
- **Attenuation Limit**:
 - *Default*: 0 dB (The AI cuts noise as much as possible).
 - *Effect*: This slider allows you to adjust the noise reduction threshold from `-100dB` to `0dB`. If you only want to reduce the noise slightly (to preserve the natural ambiance of the environment), lower this slider, for example to `-12dB` or `-24dB`. The noise suppression will then be applied more gently.

Operation:

- After tweaking the two parameters above, click the purple **Clean Selected Audio** button.
- A Progress Bar will appear, indicating that the AI is processing in the background.
- Once completed, the new crystal-clear Audio (in perfect 48kHz format) will appear right next to the original file on your computer and automatically replace the old Take on the REAPER Timeline.

3. Advanced Settings

Click on `► Advanced Settings` to expand the AI configuration panel:

- **CLI Path (AI Core Path)**:
 - Upon first launch, the Script will Auto-Detect the installation path of `deepFilter.exe` on your machine.
 - If auto-detection fails, you can manually paste the path of your `deepFilter` executable into this field.
 - *Extremely Important*: The script automatically saves this path permanently into REAPER. You only need to enter it correctly **ONCE** for a lifetime!

4. Tips & Things to Know

- **Sample Rate Security (48kHz)**: The DeepFilterNet AI core requires the input Audio to be exactly 48kHz. Don't worry if your REAPER Project is set to 44.1kHz or 96kHz. Hosi Noise Suppression features **Sample Rate Hijacking**. It will sneakily switch your Project to 48kHz in 1 millisecond to feed the audio to the AI, and then instantly revert it back to its original state. Everything is completely automatic.
- **Computer Clutter Management**: The script is programmed to 100% clean up the garbage generated during background processing (VBS Files). The exported AI Audio file will be placed in the exact original project directory instead of being dumped into the REAPER root directory.

- **Support the Developer:** If this tool saves your corrupted project, do not hesitate to click on [Donate via PayPal](#) at the bottom of the Interface!

May you achieve perfect silence in your audio world!